# MSc Advanced Project

University of the Arts London

Creative Computing Institute

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# LocalAI Analyser

Local AI for Personalisation: A Practice Based Exploration of Algorithmic Explainability and User Controlled Label Governance in Recommendations

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**Bangrui Wang**

**Supervisor:**

Prof. Mick Grierson

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**Project Development Log:**

GitHub

# Abstract

Personalised systems are increasingly affecting the way users interact with digital content. However, mainstream personalised systems often lack transparency and effective user control, resulting in a decrease in user trust. This research has developed a user-editable personalised recommendation prototype called "LocalAI Analyser". The prototype performs complete browsing history analysis locally on the device without relying on the cloud. This study aims to explore whether improving system transparency and users' control over inferred interest labels can enhance users' trust and acceptance of algorithm personalisation. The study employs a practice-centred and experiment-driven development method, combined with iterative system prototype design, task-based useability testing and structured questionnaire to evaluate the interpretability, autonomy and user experience of the system. Unlike the mainstream recommendation platform that relies on centralised data collection, this project takes editable label management as the core interaction layer, allowing users to intervene, correct or improve system inferences. The research results show that clear and visible data flow and editable interest labels can significantly improve user trust and alleviate privacy concerns. This project proposes a reusable local-first personalised system design path, emphasising the key role of privacy protection and user-autonomous label management in the recommendation system.

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# 1. Introduction

In today's digital media ecosystem, the personalised recommendation system has been widely used in all walks of life (Raza et al., 2026). As users' online time increases, the platform will analyse their browsing history to provide personalised recommendations, to improve content relevance and improve the overall user experience (Eg et al., 2023). The modern recommendation system has developed from an auxiliary function to the basic technical architecture to support the operation of the platform. Large-scale data processing and predictive modelling technology has been widely used in search ranking and news push and is increasingly changing the way users obtain information (Bojic, 2024).

However, as the personalisation mechanism is increasingly integrated into business and media platforms, people are paying more attention to its opacity and dependence on massive data (Wong et al., 2023). Users often fail to understand the generation of recommended results and lack a clear understanding of the system's reasoning, ranking logic and the scope of data collection (Govea et al., 2024). The way the platform processes user data is usually difficult to understand, which may weaken users' trust in the recommendation system and raise concerns about potential privacy violations (Sun et al., 2024).

With the progress of device-side computing and distributed technology, the method of applying language models to local data processing is attracting more and more attention. These methods emphasise data analysis and reasoning directly on the user's device, thus reducing dependence on cloud infrastructure and reducing potential privacy risks (Ubale, 2025). This method not only eliminates the need for external data transmission at the technical level but also opens up new possibilities for building a more transparent and controllable personalised system. However, there is still a lack of a personalised recommendation system that integrates device-side computing and algorithmic interpretability, which makes it difficult to assess its actual impact on user experience and user trust.

In view of this background, this study proposed and developed LocalAI Analyser, a personalised analysis system that runs entirely on the device side. The system is realised by connecting the browser extension to the locally deployed model. The model analyses the user's browsing history based on embedding and generates interest labels. Users can then directly

view and edit the reasoning results of the system. In addition, the system enhances transparency and interpretability through the logic behind the public data processing workflow and personalised recommendations. Through user testing and quantitative surveys, this study explores how local data processing and editable interest models affect users' trust, control and privacy-related concerns.

# 2. Literature Review

## 2.1 Integrated Applications of Personalized Systems

The expansion of internet infrastructure and the widespread adoption of large-scale behavioural tracking technologies have fundamentally reshaped the operation of digital advertising. Targeted digital advertising has become a core economic mechanism of the contemporary internet (OECD, 2020). With the continuous advancement of tracking capabilities and data processing technologies, advertising practices have evolved from mass-market distribution models to precision targeting based on detailed user profiles (Boerman, Kruikemeier & Zuiderveen Borgesius, 2017). Studies consistently show that behaviourally targeted advertising achieves higher relevance and conversion rates than non-targeted approaches, it makes personalized content delivery a central component of major platforms' business models (PUBLIC FIRST,2025). The advantages of personalised content have led to its widespread deployment, embedding algorithmic personalisation deeply within everyday digital interactions. At the same time, the large-scale data processing methods underpinning personalised services have heightened user concerns regarding data tracking and the scope of data usage (WEF, 2017).

## 2.2 Privacy Concerns Arising from Algorithmic Transparency Deficits

Digital media have become deeply embedded in everyday life, improving access to information, communication practices, and work efficiency. However, users generally lack an understanding of how their personal data are collected, analysed, and utilised (Loi, Ferrario & Viganò, 2021). Platforms rely extensively on users' behavioural data to drive content distribution and commercial decision-making, yet they do not provide users with a clear understanding of how their "digital profiles" or "derived data" are constructed (WEF, 2017). Establishing clear data governance frameworks, improving transparency are therefore essential for ensuring the long-term sustainability of the digital media ecosystem. Regulatory oversight of personalised advertising and scrutiny of cross-platform user data usage are also becoming increasingly stringent at this stage. The UK Competition and Markets Authority (CMA) has introduced measures such as data separation (or data silos) and data access interventions to restrict large platforms from leveraging their data advantages in one market to influence competition in others (CMA, 2020). Regulatory initiatives increasingly require platforms to

strengthen transparency and accountability to enhance public trust in personalised systems. Government guidance emphasises that embedding privacy-protective principles into system design and disclosing data practices to users are essential conditions for improving trust and ensuring the legitimacy of such systems (Government Digital Service & Central Digital and Data Office, 2021).

## 2.3 User Agency in personalized Recommendation Systems

However, the academic literature widely recognises that transparency alone is insufficient to resolve issues of trust. Even when systems disclose the data they use and the logic underlying their operations, biases may persist, and unfair outcomes may arise for certain groups if users are unable to intervene in, or correct, the system's inferences (Novelli, Taddeo & Floridi, 2024). Consequently, user agency has emerged as a critical dimension in discussions of trustworthy personalisation. Research indicates that users not only wish to understand how recommendations are generated, but also seek the ability to participate in, and even modify, the system's modelling of their interests and preferences, thereby avoiding a passive reliance on algorithmic inferences. Eslami et al.'s research shows that most users were previously unaware that platforms hid certain content, and when first confronted with evidence of such "algorithmic filtering," they commonly reacted with surprise or frustration. However, when provided with visualisation and intervention tools such as FeedVis, users exhibited a clear willingness to actively edit and manage their feeds, adjusting algorithmic inferences and redefining their personalised preferences (Eslami et al., 2015). Consequently, both industry and academia increasingly emphasise the importance of editable interest models. They contend that enabling users to view and modify the system's inferential outcomes effectively enhances user autonomy, thereby contributing to the construction of a more trustworthy personalised ecosystem (Fink, Newman & Haran, 2024).

## 2.4 Current Industry Landscape

In recent years, rising concerns over privacy breaches and behavioural tracking have prompted the industry to explore technical approaches for delivering personalised recommendations without relying on large-scale cross-site data collection. One of the most significant measures in this shift has been the restriction of cross-site tracking. Major web browsers have introduced structural limitations on third-party cookies to curb such tracking practices. Safari, for example,

blocks cross-site tracking by default through its Intelligent Tracking Prevention (ITP) mechanism (Apple Inc., 2019). This shift has substantially reduced the traditional advertising ecosystem's ability to obtain cross-platform behavioral data.

After traditional tracking methods became increasingly restricted, the industry began exploring new privacy-preserving personalisation frameworks. For example, Google's "Privacy Sandbox" (2025) aims to enable personalised inferences while reducing the use of identifiable data, employing advertising labels to deliver personalised recommendations. Some privacy focused browsers, such as Brave, have commenced exploring methods for performing inference at the device level, aiming to utilize local models to infer user interests rather than through centralised data collection. (Brave Software, 2021). Together, these technologies form an early ecosystem of privacy preserving personalization, aiming to reduce monitoring capabilities while retaining as much of the personalized user experience as possible.

## 2.5 Research Gap

Although the industry has been trying to develop personalised ways to protect privacy, existing systems still rely heavily on specific platforms. Even if users can disable personalised recommendations or restrict certain forms of data sharing, they cannot view or modify the underlying interest inference model. Due to the lack of an interpretable and operable inference process, users can only passively accept and cannot verify the accuracy of the inferred interest characteristics. This leads to a gradual decline in users' acceptance of personalised systems. To fill this research gap, we proposed and developed LocalAI Analyser. The personalised system runs entirely on the user's device and generates interest labels through embedded analysis, while allowing users to directly view, edit and manage inferred results. The system provides a new design scheme for a more transparent and controllable personalised recommendation mechanism.

# 3. Methodology

In order to meet the challenges of the current personalised recommendation system user experience, the researcher adopted a system development strategy based on practice and experiment (Candy, 2006; Zimmerman et al., 2007). Through iterative design based on user testing, and finally public user evaluation, the study examines how the editable personalisation mechanism and the transparency of data processing affect the user's trust, sense of control and overall experience.

The design of the research methodology is based on the existing literature. Previous studies have shown that the lack of algorithm transparency will significantly damage users' understanding of the reasoning process and reduce trust (Loi, Ferrario & Viganò, 2021). Research in the fields of algorithm interpretability and user control also shows that users not only attach importance to understanding algorithmic behaviour, but also attach importance to the ability to intervene, adjust or even modify the system's inference output. This kind of user autonomy has been proven to improve trust and satisfaction (Fink, Newman and Haran, 2024). These studies provide a theoretical basis for the system design logic, interaction strategy and evaluation framework adopted in this study. On this basis, the research has carried out practical system development and user testing to further explore the actual impact of these mechanisms in the localised personalised environment.

## The research is structured across three sequential phases:

**Phase 1: Preliminary Research**

This phase will combine industry analysis with semi-structured interviews to explore users' experiences with existing personalised recommendation systems, with particular focus on their perceptions regarding data privacy and user autonomy. The findings will identify core system requirements and design principles, laying the groundwork for subsequent system planning stages.

**Phase 2: System Development and Iterative Testing**

In the development stage, the experimental-driven system development method is adopted. In the process of building a personalised analysis system, through continuous small-scale prototype testing to collect user feedback and support iterative improvement. Each iteration

includes testing new features, evaluating user experience, collecting feedback, and gradually optimising the system.

**Phase 3: User Study and Evaluation**

After the main functions of the system were determined, a task-based user research and quantitative questionnaire were carried out. The evaluation examines the differences in user experience between the existing personalised systems and the proposed systems, paying special attention to issues related to transparency, personalised control and privacy. The goal is to systematically evaluate whether the editable device-side personalisation system can improve user experience and improve users' acceptance of personalised recommendations in actual use scenarios.

# 4. System Development

## 4.1 Rethinking Personalized Systems

Although major platforms have developed mature recommendation architectures, as illustrated in Figure 1, these systems typically rely on the collection of substantial user data, which is then analysed according to the system's pre-set algorithmic logic (Eckhardt, 2009). Users cannot modify either the personalisation process or its outcomes, and they have limited means of determining whether their data are being handled safely. Such “black box” personalisation imposes clear constraints on transparency and privacy control. Figure 2 illustrates Google Chrome’s interface for managing personalized interest categories through the Ad Topics settings(Google, 2025）. However, users still cannot intuitively view the label structure behind the recommendation, nor can they intervene or modify the reasoning process of the system. In order to overcome these limitations, the system developed by this research has broken through the traditional personalised practice. By implementing a fully localised and controllable data analysis process and generating editable interest labels, the design explores an alternative path that is completely different from the current platform-centered personalisation model.

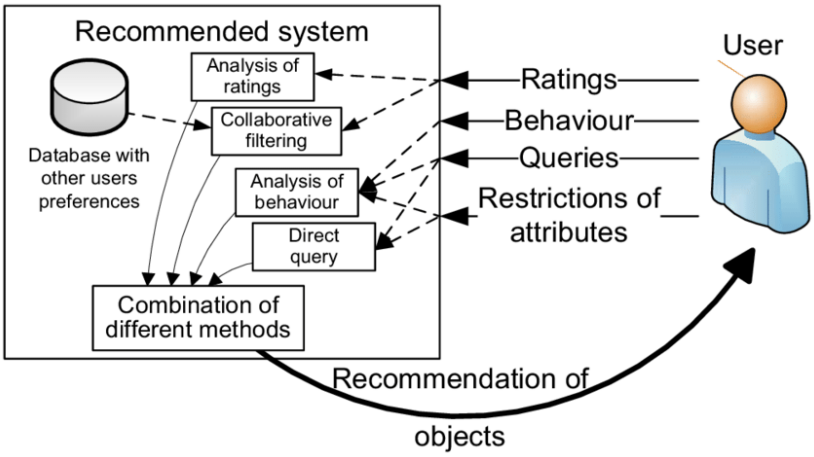


Figure 1: Structure of a recommender system. Source: Eckhardt (2009).

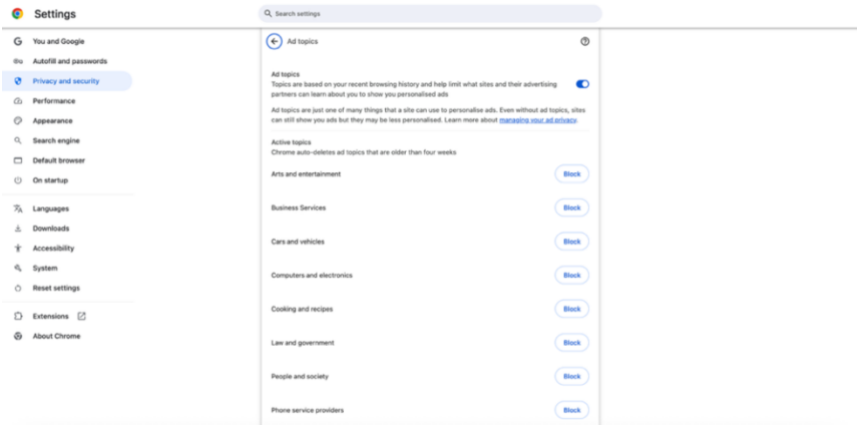


Figure 2: Google Ad topics

## 4.2 System Architecture Design

The system adopts an architecture in which the browser extension works together with the local analysis back-end, to fully personalise the device. Unlike the mainstream platform structure described in the previous section, this architecture is designed to provide user-controlled personalised processes without relying on external servers.

As shown in Figure 3, the browser extension serves as the interactive layer of the system, allowing users to initiate analysis, inspect the data included in processing, and adjust the resulting interest labels. The local backend performs interest-label inference and personalized content generation. All data processing and storage remain strictly local to the device with no cloud services are contacted, nor are external data copies created. This separation of design between the interaction layer and the analytics layer ensures system usability while reducing the risk of data leakage, thereby enhancing user trust in the system.

By localising the entire data processing workflow, we circumvent the opaque reasoning pathways inherent in traditional platforms, establishing a user-centric personalised recommendation system. Under this model, personalisation ceases to be an operation

performed by the platform on behalf of the user, becoming instead a process actively controlled by the user on their own device.

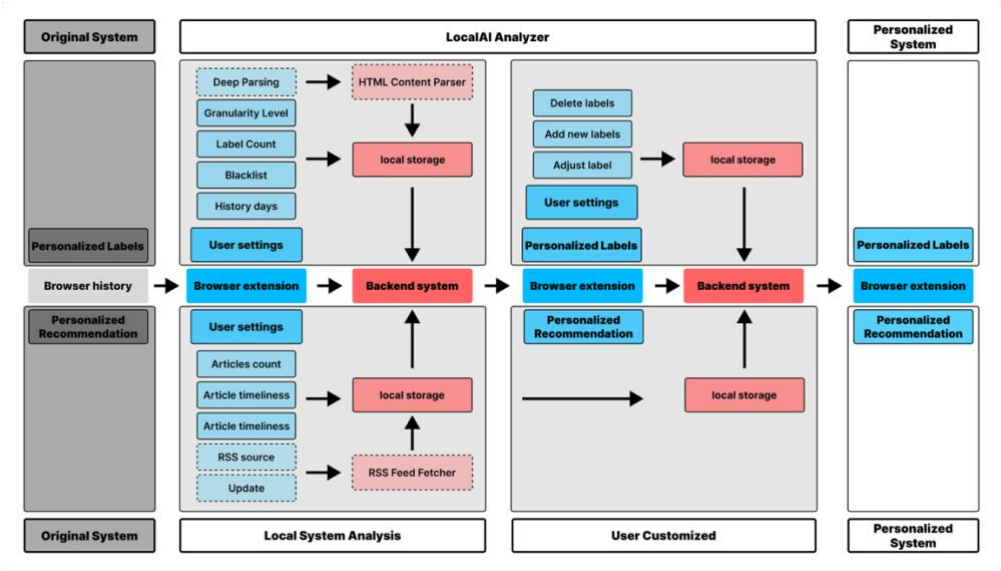


Figure 3: Locally Editable Personalised Systems

## 4.3 Models Used in the System

The system employs a locally executed semantic embedding model (Sentence Transformer: all-mpnet-base-v2) as the core component for analysing browsing history entries and generating personalised content. Instead of relying on external APIs or cloud-based inference services, all text processing and vector computation are performed entirely on the user's device. Webpage titles, descriptions, and extracted textual content are transformed into embedding vectors that capture their semantic characteristics. To infer the most relevant interest category for each record, the system calculates the cosine similarity between the vector representation of a history entry and the vectors of predefined label descriptions. Cosine similarity provides a standard measure of semantic proximity between two vectors and is computed as:

$$\text{cosine similarity} = S_C(A,B) \coloneqq \cos(\theta) = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\cdot\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Let H denote the embedding vector of a browsing-history entry and L the vector representing a label definition. The system assigns the label with the highest similarity score:

$$\text{score}(H, L) = \frac{H \cdot L}{\| H \| \| L \|}$$

As illustrated in Figure 4, the inferred labels are subsequently aggregated and weighted to form the final interest profile presented to the user. Since all computations take place on the local device, the embedding model serves as the foundational mechanism enabling the system's on-device personalisation workflow.


```
  "data": [
    {
      "url": "https://github.com/InteractiveAdvertisingBureau/iabtcf-es",
      "title": "InteractiveAdvertisingBureau/iabtcf-es: Official compliant tool suite for implementing the Transparency and Consent
Framework (TCF) v2.0. The essential toolkit for CMPs.",
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Consent Framework (TCF) v2.0. The essential toolkit for CMPs. - Official compliant tool suite for implementing the Transparency and
Consent Framework (TCF) v2.0. The essential toolkit for CMPs. - InteractiveAdvertisingBureau/iabtcf-es",
      "matches": [
        {
          "score": 0.3294,
          "path": "Sensitive Topics > Online Piracy"
        },
        {
          "score": 0.3258,
          "path": "Technology & Computing > Computing > Information and Network Security"
        },
        {
          "score": 0.3213,
          "path": "Sensitive Topics > Adult & Explicit Sexual Content"
        },
        {
          "score": 0.3172,
          "path": "Sensitive Topics > Crime & Harmful Acts to Individuals, Society & Human Right Violations"
        },
        {
          "score": 0.3115,
          "path": "Sensitive Topics > Illegal Drugs, Tobacco, eCigarettes, Vaping, Alcohol"
        }
      ]
    },
```


Figure 4: Label Similarity Scores

## 4.4 Local Data Flow

The system is built on the principles of data transparency and user control, so that users can directly understand the source, purpose and operation logic of each stage of the personalised workflow. Unlike mainstream platforms that automatically collect and process data in a closed environment, all analysis operations in LocalAI Analyser are explicitly initiated by users. Users can clearly see the data included in the analysis, the start time of the analysis, and the stage the system is currently processing. The system will not retrieve or transmit any information in the background. All data streams are displayed through the real-time output of the interface indicator and the local terminal, ensuring full traceability. During the back-end processing process, the terminal will continuously update the operating status of the system (as shown in Figure 5), so that users can monitor the progress of the system at any time. In addition to data visibility, users also have the right to decide whether data enters the analysis process, whether it is stored, and when personalised content is updated. This design directly responds to the

concerns in the literature about the decline in trust due to the opaque algorithm process. By limiting all calculations to local devices, the system replaces the non-transparent and auditable reasoning mechanism of the traditional platform. It gives users the right to supervise and control the entire life cycle of their data, restores the interpretability and autonomy of the personalised process, and transforms users from passive data sources to active participants in system operation.

```
whisky — LocalAI_analyse — LocalAI_analyse ◂ LocalAI_analyse — 80×24
tory/history_1764213108044.json → /Users/whisky/Desktop/Extension_Test_02/backen
d/dist/LocalAI_analyse/history_exports/history_latest.json
[Setting] Analysis parameters received: {'historyDays': 30, 'granularityLevel':
1, 'samplingCount': 20, 'siteBlacklist': [], 'useDeepParsing': False}
==================================================================
[Analysis] Starting full analysis pipeline...
[Analysis] Running history data analysis (embedding, clustering, and label extra
ction)...
[Setting] Setting: deepParsing=False, TOP_N=5, THRESHOLD=0.39, granularityLevel=
1, samplingCount=20, blacklistCount=0
[Setting] Blacklist filtering completed: 1842 / 1842 records retained.
[BeautifulSoup] Deep parsing:false
[Model] Loading model and taxonomy library...
[Model] Taxonomy loaded successfully with 703 entries.
[File] Embedding comparison analysis file exported: embedding_analysis.json
[File] custom_analysis_result.json updated.
[Analysis] Historical data analysis completed. Generated 20 user interest tags.
==================================
[RSS] rss_summary.json not found — starting RSS fetch process.
[RSS] Starting FAST multi-threaded RSS fetching...
[Setting] Configuration loaded: {'enabled': True, 'feeds': [], 'historyDays': 19
, 'recommendCount': 100, 'autoUpdate': True, 'updateIntervalHours': 1}
[RSS] Only keeping articles newer than 19 days.
[RSS][OK] TechCrunch: 20 items
```

Figure5: Visualised Analysis Workflow

## 4.5 User Interface Design

The interface is designed to strengthen the user's ability to influence the personalisation process. In many existing personalised systems, users have little awareness of how their data is selected or interpreted, and even less opportunity to adjust the outcomes. This system takes a different approach by allowing users to decide which records are included in the analysis, how detailed the resulting interest labels should be, and how the automatically generated categories should be modified. These points of adjustment give users direct control over how their browsing behaviour is translated into personalised outputs. As shown in Figure 6, users can control the scope of data that enters the analysis and select the level of precision for the generated interest labels. The system also provides a way for users to maintain their own set of content sources for recommendations, enabling them to build a personalised library rather than relying on predefined data inputs from external platforms. In addition, the interface supports the correction and refinement of system-generated interest labels, as illustrated in Figure 7. Users

may remove categories that do not reflect their preferences, revise inaccurate assignments, or add new labels that express interests not captured by the automated process. Together, these features reposition the user within the personalisation workflow and enable a more active, continuous form of participation, addressing long-standing concerns about limited transparency and restricted user autonomy in personalised systems.

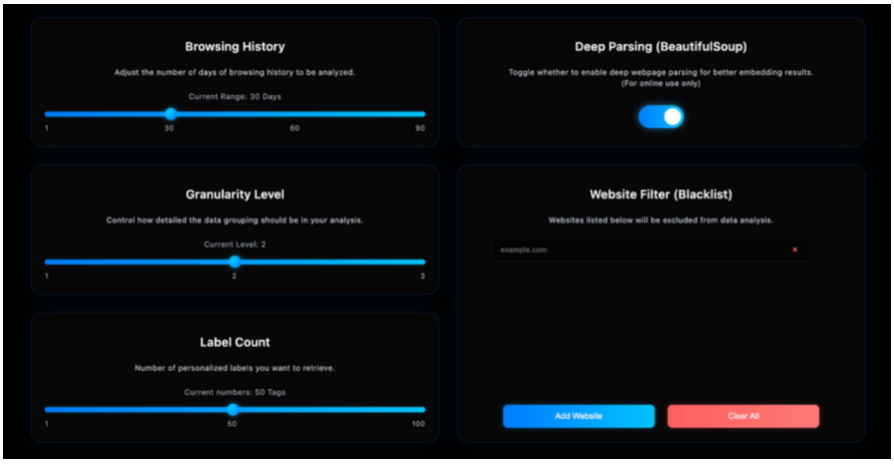



Figure6: User Control

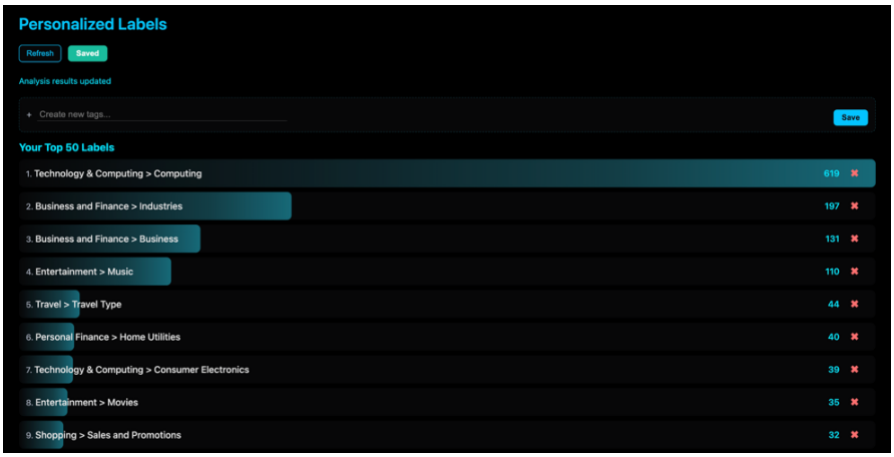



Figure7: Personalised Label

## 4.6 Iterative Development

In the process of developing LocalAI Analyser, an experiment-driven system development strategy was adopted to continuously improve the system through continuous technical verification and user feedback. Each iteration is based on the insights obtained in the previous stage, and the system has also developed from an early conceptual prototype to a stable, user-

centred, device-built personalised tool. The following will summarise the three main iterations and their contributions to the maturity of the system.

**Iteration 1: Local Embedding and Label Inference Feasibility**

In the first round of iteration, it was verified that embedded calculations and labelling inferences could be reliably completed on local devices. By using the browsing history of three users, the system successfully generated a preliminary interest label, which confirmed the technical feasibility of device-side vector processing. This step lays the necessary foundation for building a fully localised personalised workflow.

**Iteration 2: System Integration and Editable Labels**

In the second version, a complete data pipeline is built by connecting the browser extension to the local backend. The editable label function is introduced, which allows users to modify the analysed data and inferred categories. Small-scale tests show that editable labels improve users' understanding and sense of control.

**Iteration 3: Interface Refinement and Enhanced User Controls**

In the third update, user feedback was adopted, the interface was improved, more customisation options were added, and the operation process was simplified. In general, this update makes system control more intuitive while maintaining functional integrity and provides users with a more convenient way to manage and adjust relevant settings.

After these three iterations, LocalAI Analyser has developed from an early feasibility prototype to a fully integrated, user-centred system that can provide a personalised experience on the device. After the core functions of the system are stable, the research enters the next stage: evaluate the availability in a real interactive environment and collect user feedback through task-based experiments and structured surveys, focussing on the user's experience, understanding and overall response to the system.

# 5. User Test

## 5.1 Research Questions

The evaluation in this study is structured around the following three research questions:

**RQ1:** Does on-device analysis increase users' trust in personalized systems and reduce concerns about potential privacy breaches?

**RQ2:** Do editable interest labels enhance users' sense of control (user agency) within the system?

**RQ3:** Does the system's interface help users develop a clearer understanding of the personalisation process?

### 5.2 Research Procedure

In order to assess the useability of the system and the clarity of the operation process, participants are required to perform a series of short operations to simulate their expected use of the system. They first export a set of browsing records and then start the local analysis process in the interface. After the analysis is completed, the participants review the interest labels automatically generated by the system and adjust if necessary, and then verify the personalised content generated by the system based on their edited profile.

Throughout the experiment, the researchers observed how the participants completed these steps and recorded the moments of their hesitation, confusion or smooth interaction, thus initially constructing a qualitative image of useability. After the task, the participants completed a two-part questionnaire. The first part focuses on their previous experience of using mainstream personalised systems, and the second part asks how their perception of transparency, control and privacy has changed after using LocalAI Analyser. By comparing the answers before and after the task, researchers can have a clearer understanding of how the system mechanism affects their overall experience.

## 5.3 Participant Information

A total of 16 participants took part in the study. All were regular users of personalized recommendation systems and included students, creative computing practitioners, and

general internet users. Participants were recruited through open calls on campus and voluntary responses via social media.

As shown in Figure 8, 92.3% of participants reported using personalisation features in their daily online activities, while 76.9% expressed concerns about privacy leakage. These findings provide a concrete rationale for examining the need for transparency and localized processing in personalized systems.

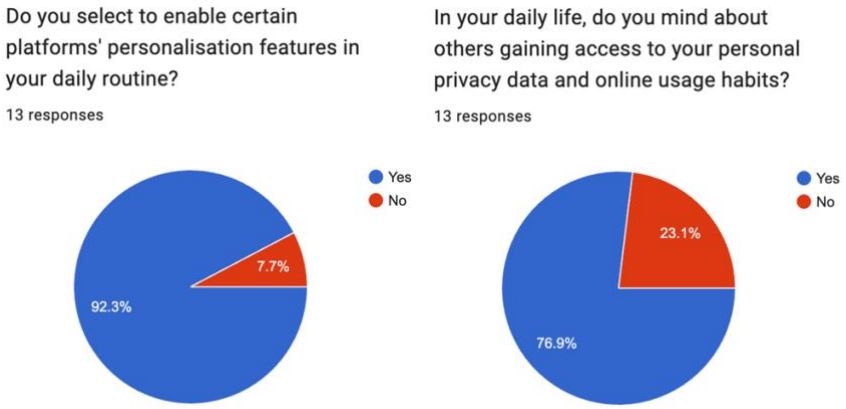
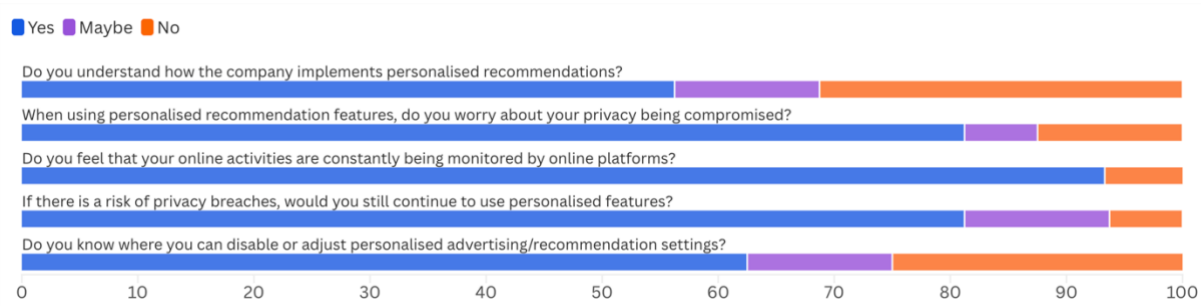



Figure8: Participant Information

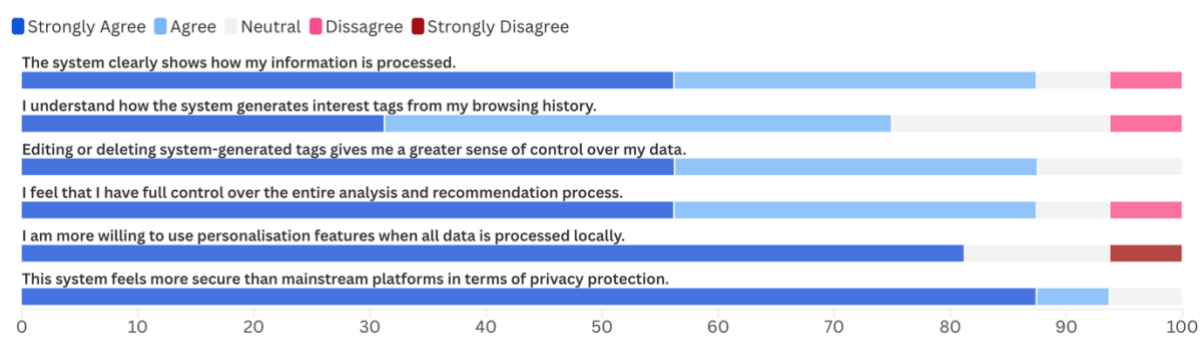
## 5.4 Results

### 5.4.1: Pre-test Perceptions




Although participants said that personalised features were often used, many did not understand how these systems worked and expressed serious concerns about privacy and trust issues. Most people don't understand how recommendations are generated and don't know how to adjust the relevant settings, which highlights the need to improve transparency and user control.

### 5.4.2 Post-test Experience




After using the system, participants said that they had a deeper understanding of how their data was processed and could edit the labels generated by the system, thus having greater control over their data. Many participants also said that their privacy concerns had been alleviated. In general, participants' perception of transparency and security has improved significantly after using the system.

## 5.5 Limitations of the User Study

Although this study provides valuable insights into how users interact with the system, there are still some limitations. The sample size is relatively small, and the participants only use the device for a short time, which may not fully reflect the long-term use pattern. In addition, the evaluation is based on the participant's self-reported impression rather than behavioural data, which may affect the accuracy of the research results. Finally, since this study describes the system as a fully localised solution, the trust response of participants may be affected by this framework. These factors must be considered when interpreting the research results.

# 6. Discussion

This chapter discusses the impact of the system on transparency, control, and privacy experience based on the results of user tests. The overall trend shows that after using the system, participants have significantly improved their understanding of personalized processes, their level of participation, and their sense of data security.

## 6.1 Interpreting the Findings

User test results show that before using the system, most participants had low trust in mainstream personalised platforms and were concerned about data collection and interest category generation. After completing the task sequence, the participants had a significant and clearer understanding of the data processing process and the logic of interest labelling inference. The fully localised processing process was particularly well received. Many participants said that device-side data analysis enhanced their confidence in system reliability and reduced their concerns about data leakage. These results show that localisation is not only a technical decision, but also an important source of psychological comfort for users. The editable label system has further changed the role of users in the process of personalisation. Traditionally, users are passive recipients of algorithm results. In contrast, the current system allows users to correct wrong inferences, add new interests, and delete unnecessary categories. This intervention ability enhances the user's sense of control and transforms personalised recommendations from passive acceptance of calculation results into an interactive process. Participants generally believed that this flexibility improved the acceptability of the system and met their expectations of personalised technology.

## 6.2 Comparison with Existing Personalisation Models

The structure proposed in this study is completely different from the existing business model. Mainstream systems usually process data remotely, often do not disclose the reasons for their output results, and do not provide a mechanism for users to modify these characteristics. The model proposed in this study challenges this paradigm, which realises data analysis through the localisation model, and makes the data flow and generated labels visible and editable. This method transforms the personalised system from a platform-controlled process to a process that prioritises user control. Feedback from participants further emphasised this difference, and many pointed out that locally editable systems are more acceptable than traditional calculation

methods. This shows that people's distrust of platform-based personalised technology stems from the lack of interpretation mechanism and user autonomy, not from personalisation itself.

## 6.3 Relation to Prior Literature

Our user test results are highly consistent with existing research. Research on transparency and algorithmic interpretability shows that an opaque reasoning process will reduce users' trust and understanding. User tests show that participants can identify the reasoning logic behind the system output through the interface, which proves that the visualised data flow can enhance users' understanding of algorithm logic and significantly improve their trust in the system. Research on user autonomy further shows that being able to intervene in algorithmic decision-making can improve the overall experience. The editable label design in the LocalAI analyser also confirms this principle. By actively participating in system operation with custom settings, users can change from passive data providers to active contributors to the recommendation process. Our test results show that improving transparency and control significantly improves users' acceptance of the system, which is highly consistent with the theoretical discussion of algorithm interpretability and user autonomy.

## 6.4 Future Work

Based on the findings of this study, the future research direction is gradually clear. First of all, this user study adopts a short task-oriented form, so we have a limited understanding of how users interact with the system over time. Conducting longitudinal research will help a more comprehensive understanding of the changes in trust, perception control and editable personalised value of users when using LocalAI analysers in daily activities. Secondly, while participants expressed appreciation for the fully localised workflow, many expressed their desire to continue using the system outside the main platform or application. Therefore, future research can explore how to integrate locally generated interest models with external services under the premise of strictly dividing the boundaries of device-side computing and data. This method will enable the system to provide richer and more diverse sources of recommendation without compromising user privacy. In addition, integrating the system into the real browser ecosystem can more realistically evaluate its convenience and stability in daily use. This will lay a more solid foundation for the development of a truly controllable, transparent and privacy-protecting personalised system.

# 7. Conclusion

This study explores a new personalised system approach, which emphasises user-centred design and privacy protection. By developing a localised and editable personalised system, we provide a solution to replace the platform-driven personalised model and show how the personalised system gives users the right to participate while maintaining transparency and interpretability.

Research shows that personalised recommendations do not need to rely on remote servers or complex reasoning processes. The locally run analysis model allows users to directly observe the data processing and label generation process and modify it if necessary. This transparent and adjustable mechanism significantly improves the user's trust in the system and changes the user's role in the process of personalisation from passive acceptance to active participation.

More broadly, the project shows how artificial intelligence systems can reduce dependence on centralised data infrastructure while improving user autonomy, providing a model for the future development of personalised tools, which can ensure user control and privacy without affecting functions.

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